AS 2 · FD01

Bit

AS 2 · FD01

Nibble

AS 2 · FD01

Byte

AS 2 · FD01

Kilobyte (KB)

AS 2 · FD01

Megabyte (MB)

AS 2 · FD01

Gigabyte (GB)

AS 2 · FD01

Terabyte (TB)

AS 2 · FD01

Place value

Backs · print double-sided, flip on long edge

A group of 4 bits; half a byte, represented by one hexadecimal digit. Nibble	A binary digit; the smallest unit of data, which stores either 0 or 1. Bit
1024 (2^{10}) bytes. Kilobyte (KB)	A group of 8 bits, which can represent 256 different values. Byte
1024 megabytes, or 2^{30} bytes. Gigabyte (GB)	1024 kilobytes, or 2^{20} (1 048 576) bytes. Megabyte (MB)
The value of a column in a number, determined by its position and the base, e.g. 1, 2, 4, 8 in binary. Place value	1024 gigabytes, or 2^{40} bytes. Terabyte (TB)

AS 2 · FD01

Most significant bit (MSB)

AS 2 · FD01

Least significant bit (LSB)

AS 2 · FD01

Hexadecimal

AS 2 · FD01

Binary-coded decimal (BCD)

AS 2 · FD02

Binary addition

AS 2 · FD02

Carry

AS 2 · FD02

Sign bit

AS 2 · FD02

Sign and magnitude

Backs · print double-sided, flip on long edge

The rightmost bit of a binary number, with a place value of 1. Least significant bit (LSB)	The leftmost bit of a binary number, which has the highest place value. Most significant bit (MSB)
A representation in which each denary digit is stored separately as its own 4-bit binary code. Binary-coded decimal (BCD)	A base-16 number system using the digits 0–9 and A–F, where each digit represents one nibble (4 bits). Hexadecimal
A bit passed to the next column on the left when a column total is two or more. Carry	Adding binary numbers column by column from the right using the rules 0+0=0, 0+1=1, 1+1=0 carry 1, 1+1+1=1 carry 1. Binary addition
A signed-integer representation in which the MSB indicates the sign (0 positive, 1 negative) and the remaining bits give the size. Sign and magnitude	The most significant bit of a signed number, used to show whether the value is positive or negative. Sign bit

AS 2 · FD02

Two's complement

AS 2 · FD02

Unsigned integer

AS 2 · FD02

Range (integers)

AS 2 · FD02

Overflow

AS 2 · FD02

Overflow flag

AS 2 · FD03

Real number

AS 2 · FD03

Fixed-point representation

AS 2 · FD03

Binary point

Backs · print double-sided, flip on long edge

A whole number stored with no sign bit, so all bits represent magnitude and only zero or positive values are possible.

Unsigned integer

A signed-integer representation in which the MSB has a negative place value; a number is negated by inverting its bits and adding 1.

Two's complement

An error that occurs when the result of a calculation is too large or too small to be stored in the available number of bits.

Overflow

The smallest and largest values a representation can store with a given number of bits, e.g. −128 to +127 in 8-bit two's complement.

Range (integers)

A number that can have a fractional part, such as 6.75 or −0.3125.

Real number

A bit in the processor's status register that is set when an arithmetic operation overflows.

Overflow flag

The point separating the whole-number bits from the fractional bits, whose place values are ½, ¼, ⅛ and so on.

Binary point

A way of storing real numbers in which the binary point is at an agreed, unchanging position.

Fixed-point representation

AS 2 · FD03

Floating-point representation

AS 2 · FD03

Mantissa

AS 2 · FD03

Exponent

AS 2 · FD03

Normalisation

AS 2 · FD03

Precision

AS 2 · FD03

Range (floating point)

AS 2 · FD03

Rounding error

AS 2 · FD03

Underflow

Backs · print double-sided, flip on long edge

The part of a floating-point number that holds its significant digits.

Mantissa

A way of storing real numbers as a mantissa multiplied by 2 to the power of an exponent, allowing a wide range of values.

Floating-point representation

Adjusting a floating-point number so the first two bits of the mantissa differ, maximising precision and giving a unique representation.

Normalisation

The part of a floating-point number that states how many places, and in which direction, the binary point is moved.

Exponent

The span between the largest and smallest magnitudes that can be represented, determined mainly by the exponent bits.

Range (floating point)

How finely values can be distinguished, determined by the number of significant bits (mantissa bits).

Precision

An error that occurs when a value is too close to zero to be represented with the available exponent.

Underflow

The difference between a real value and the nearest value that can be stored in the available bits.

Rounding error

AS 2 · FD04

Character set

AS 2 · FD04

ASCII

AS 2 · FD04

Extended ASCII

AS 2 · FD04

Unicode

AS 2 · FD04

UTF-8

AS 2 · FD04

Control character

AS 2 · FD04

Parity bit

AS 2 · FD04

Check digit

Backs · print double-sided, flip on long edge

American Standard Code for Information Interchange: a 7-bit character set of 128 codes covering English letters, digits, punctuation and control characters.

ASCII

An agreed list of characters that a computer system can represent, in which each character is given a unique binary code.

Character set

A character standard that gives a unique code point to characters from almost all of the world's writing systems, plus symbols and emoji; its first 128 codes match ASCII.

Unicode

An 8-bit extension of ASCII giving 256 codes; codes 0–127 match ASCII and codes 128–255 hold accented letters and symbols, which vary between code pages.

Extended ASCII

A non-printing character code, such as carriage return, line feed or tab, used to control how text is processed or displayed.

Control character

A variable-length Unicode encoding that uses one byte for ASCII characters and up to four bytes for other characters.

UTF-8

An extra digit calculated from the other digits of a code number and added to the end, so the computer can detect entry or reading errors by recalculating it.

Check digit

An extra bit added to a group of bits so that the total number of 1s is even (even parity) or odd (odd parity), allowing transmission errors to be detected.

Parity bit

AS 2 · FD04

Modulus 11

AS 2 · FD04

Transposition error

AS 2 · FD05

Pixel

AS 2 · FD05

Resolution

AS 2 · FD05

Colour depth

AS 2 · FD05

Metadata

AS 2 · FD05

Bitmap

AS 2 · FD05

Vector graphic

Backs · print double-sided, flip on long edge

An input error in which two characters are entered in the wrong order, such as 47019 instead of 40719.

Transposition error

A check digit method in which each digit is multiplied by a weighting, the products are added, and the check digit is found from the remainder after dividing by 11 (0 gives 0, 1 gives X, otherwise 11 minus the remainder).

Modulus 11

The number of pixels in an image, usually given as width × height in pixels, or as dots per inch (dpi) for printed images.

Resolution

Short for picture element: the smallest addressable element of a bitmap image, storing a single colour as a binary value.

Pixel

Data about data, such as an image's width, height and colour depth, stored with the file so that software can interpret it correctly.

Metadata

The number of bits used to represent the colour of each pixel; n bits give 2^n possible colours, and more bits give a larger file.

Colour depth

An image stored as a set of objects described by mathematical properties such as coordinates, size and colour, so it can be scaled without loss of quality.

Vector graphic

An image stored as a grid of pixels, with the colour of each pixel held as a binary number.

Bitmap

AS 2 · FD05

Sampling rate

AS 2 · FD05

Sample resolution

AS 2 · FD05

Analogue-to-digital converter (ADC)

AS 2 · FD05

Nyquist theorem

AS 2 · FD05

Pixelation

AS 2 · FD06

Data

AS 2 · FD06

Information

AS 2 · FD06

Knowledge

Backs · print double-sided, flip on long edge

The number of bits used to store each sound sample, which determines how precisely each amplitude is recorded; also called bit depth.

Sample resolution

The number of samples of an analogue sound wave taken per second, measured in hertz.

Sampling rate

The principle that the sampling rate must be at least twice the highest frequency in the signal for it to be reproduced accurately.

Nyquist theorem

A device that measures an analogue signal at regular intervals and converts each measurement into a binary value.

Analogue-to-digital converter (ADC)

Raw facts and figures, such as numbers, text, images or sound, that have no context and therefore no meaning on their own.

Data

The blocky appearance of a bitmap image when it is enlarged beyond its resolution so that individual pixels become visible.

Pixelation

The result of applying one or more rules to information so that an informed decision can be made.

Knowledge

Data that has been processed and given context so that it has meaning to the person using it.

Information

AS 2 · FD06

GIGO

AS 2 · FD06

Transposition error

AS 2 · FD06

Transcription error

AS 2 · FD06

Validation

AS 2 · FD06

Range check

AS 2 · FD06

Format check

AS 2 · FD06

Lookup check

AS 2 · FD06

Verification

Backs · print double-sided, flip on long edge

An input error in which two characters are entered in the wrong order, such as 47019 instead of 40719.

Transposition error

Garbage in, garbage out: if the input data is of poor quality, the output information will also be of poor quality.

GIGO

An automatic check carried out by software to make sure that input data is sensible and reasonable and obeys set rules.

Validation

An input error in which data is copied or typed incorrectly, such as 1500 instead of 150.

Transcription error

A validation check that data matches a set pattern of characters, such as two letters followed by four digits.

Format check

A validation check that a value lies between a lower and an upper limit.

Range check

A check that data has been entered or copied accurately, so that it exactly matches the original source.

Verification

A validation check that compares the entered value with a list or file of valid values.

Lookup check

AS 2 · FD06

Double entry

AS 2 · FD07

Check digit

AS 2 · FD07

Modulus 11

AS 2 · FD07

Batch total

AS 2 · FD07

Control total

AS 2 · FD07

Hash total

AS 2 · FD07

Batch processing

AS 2 · FD07

Real-time transaction processing

Backs · print double-sided, flip on long edge

An extra digit calculated from the other digits of a code number and added to the end, so the computer can detect entry or reading errors by recalculating it.

Check digit

A verification method in which data is entered twice and the computer compares the two versions, flagging any difference.

Double entry

A validation check in which the number of documents in a batch is counted by hand and compared with the number entered by the computer.

Batch total

A check digit method in which each digit is multiplied by a weighting, the products are added, and the check digit is found from the remainder after dividing by 11 (0 gives 0, 1 gives X, otherwise 11 minus the remainder).

Modulus 11

A batch total formed by adding up a field whose total has no meaning, such as ID numbers, and comparing it with the computer total.

Hash total

A batch total formed by adding up a meaningful field on every document, such as total hours worked, and comparing it with the computer total.

Control total

Real-time processing in which each transaction is processed as it occurs while the user waits, so the master file is always up to date, e.g. bookings and cash machines.

Real-time transaction processing

Processing in which transactions are collected over a period in a transaction file and processed together as one job at a later time, without user intervention.

Batch processing

AS 2 · FD07

Real-time processing

AS 2 · FD07

Record locking

AS 2 · FD07

Multitasking

AS 2 · FD07

Multithreading

AS 2 · FD07

Multi-user

AS 2 · FD08

Arithmetic logic unit (ALU)

AS 2 · FD08

Control unit (CU)

AS 2 · FD08

Register

Backs · print double-sided, flip on long edge

Preventing other users from changing a record while one user is updating it, to avoid lost updates in multi-user systems.

Record locking

Processing in which each transaction or input is dealt with as soon as it occurs, so the master file is always up to date; real-time control systems also guarantee a response fast enough to affect the next input.

Real-time processing

Running several threads, each a separate path of execution, within one process so they share its memory and code.

Multithreading

The apparent running of several programs at once by rapidly switching the processor between them.

Multitasking

The part of the CPU that performs arithmetic operations, logical operations and comparisons on data held in registers.

Arithmetic logic unit (ALU)

A system that allows many users to share the same computer's resources at the same time, each with their own session and permissions.

Multi-user

A single, very fast storage location inside the CPU that holds data, an instruction or an address during processing.

Register

The part of the CPU that fetches and decodes instructions and sends control signals to coordinate the processor, memory and I/O.

Control unit (CU)

AS 2 · FD08

Program counter (PC)

AS 2 · FD08

Memory address register (MAR)

AS 2 · FD08

Memory data register (MDR)

AS 2 · FD08

Current instruction register (CIR)

AS 2 · FD08

Accumulator (ACC)

AS 2 · FD08

Address bus

AS 2 · FD08

Data bus

AS 2 · FD08

I/O controller

Backs · print double-sided, flip on long edge

A register that holds the address of the memory location currently being read from or written to. Memory address register (MAR)	A register that holds the address of the next instruction to be fetched; it is incremented after each fetch. Program counter (PC)
A register that holds the instruction currently being decoded and executed. Current instruction register (CIR)	A register that holds data or an instruction just read from memory, or data waiting to be written to memory. Memory data register (MDR)
The unidirectional part of the system bus that carries the address of a memory location or I/O device from the CPU. Address bus	A register that holds the results of calculations and logical operations performed by the ALU. Accumulator (ACC)
An electronic circuit between the system bus and a peripheral that converts protocols, voltages and data formats and buffers data. I/O controller	The bidirectional part of the system bus that carries data and instructions between the CPU, memory and I/O devices. Data bus

AS 2 · FD08

Von Neumann architecture

AS 2 · FD09

Fetch–decode–execute cycle

AS 2 · FD09

Opcode

AS 2 · FD09

Operand

AS 2 · FD09

Instruction set

AS 2 · FD09

Pipelining

AS 2 · FD09

Clock speed

AS 2 · FD09

Core

Backs · print double-sided, flip on long edge

The continuous process by which the CPU fetches an instruction from memory, decodes it and carries it out.

Fetch–decode–execute cycle

A computer design in which instructions and data are stored together in the same memory and accessed by a single processor over a shared bus.

Von Neumann architecture

The part of a machine-code instruction that holds the data to be used or the address where the data is stored.

Operand

The part of a machine-code instruction that specifies the operation to be performed.

Opcode

Overlapping the fetch, decode and execute stages of successive instructions so that several instructions are being processed at once.

Pipelining

The complete set of machine-code instructions that a particular processor is designed to recognise and execute.

Instruction set

An independent processing unit within a CPU, able to fetch, decode and execute its own instructions.

Core

The number of clock cycles the processor carries out per second, measured in hertz (e.g. 3.5 GHz).

Clock speed

AS 2 · FD09

Multi-core processor

AS 2 · FD09

Word size

AS 2 · FD09

Cache

AS 2 · FD09

Benchmark

AS 2 · FD09

FLOPS

AS 2 · FD10

Main memory

AS 2 · FD10

RAM

AS 2 · FD10

ROM

Backs · print double-sided, flip on long edge

The number of bits a processor can handle as a single unit in one operation, usually equal to the size of its registers. Word size	A single processor chip containing two or more independent cores that can execute instructions at the same time. Multi-core processor
A standard set of tasks run on different computers so that their performance can be measured and compared fairly. Benchmark	Small, very fast memory close to the CPU that stores frequently used instructions and data to reduce slower accesses to RAM. Cache
The memory that the processor can address directly, holding the programs and data currently in use (RAM and ROM). Main memory	Floating-point operations per second: a measure of a processor's speed at arithmetic on real numbers. FLOPS
Read-only memory: non-volatile main memory whose contents are fixed when made, used for start-up firmware and embedded programs. ROM	Random access memory: volatile read/write main memory that holds the operating system, programs and data currently in use. RAM

AS 2 · FD10

Volatile

AS 2 · FD10

Cache memory

AS 2 · FD10

Cache controller

AS 2 · FD10

Virtual memory

AS 2 · FD10

Disk thrashing

AS 2 · FD10

Secondary storage

AS 2 · FD10

Serial access

AS 2 · FD10

Optical storage

Backs · print double-sided, flip on long edge

Small, very fast and expensive memory on or beside the processor that holds frequently used data and instructions and data read ahead from RAM.

Cache memory

Describes memory that loses its contents when the power is switched off.

Volatile

An area of secondary storage used by the operating system as an extension of RAM when RAM is full, with pages swapped in and out.

Virtual memory

The device that transfers data from RAM into cache, predicting and pre-fetching the data the processor is likely to need next.

Cache controller

Non-volatile storage that keeps programs and data permanently when the power is off, e.g. hard disk, SSD, optical disc.

Secondary storage

Severe loss of performance when the computer spends more time swapping pages between RAM and virtual memory than processing.

Disk thrashing

Storage in which a laser writes and reads marks on the reflective surface of a disc, e.g. CD, DVD and Blu-ray.

Optical storage

Data must be read in order from the start until the required item is reached, as on magnetic tape.

Serial access

AS 2 · FD10

Solid-state storage

AS 2 · FD11

Parallel processing

AS 2 · FD11

Multi-core processor

AS 2 · FD11

Thread

AS 2 · FD11

Harvard architecture

AS 2 · FD11

Embedded system

AS 2 · FD11

RISC

AS 2 · FD11

CISC

Backs · print double-sided, flip on long edge

The simultaneous use of two or more processors or cores to work on different parts of the same task.

Parallel processing

Non-volatile storage using flash memory with no moving parts, e.g. SSDs, USB drives and memory cards.

Solid-state storage

A sequence of instructions within a program that can be scheduled to run independently, for example on a separate core.

Thread

A single chip containing two or more complete processing cores that can each execute instructions at the same time.

Multi-core processor

A computer system built into a larger device to perform one dedicated function, such as a heating controller.

Embedded system

A computer design with separate memories and buses for instructions and data, so both can be accessed at the same time.

Harvard architecture

Complex instruction set computer: a processor with a large set of instructions, some of which carry out several low-level operations and take several cycles.

CISC

Reduced instruction set computer: a processor with a small set of simple, fixed-length instructions, most executed in one clock cycle.

RISC

AS 2 · FD11

Graphics processing unit (GPU)

AS 2 · FD11

SIMD

AS 2 · FD11

Co-processor

AS 2 · FD11

Pipelining

AS 2 · FD12

User interface

AS 2 · FD12

Graphical user interface (GUI)

AS 2 · FD12

Command line interface (CLI)

AS 2 · FD12

Forms dialogue interface

Backs · print double-sided, flip on long edge

Single instruction, multiple data: one instruction is applied to many data items at the same time. SIMD	A processor with many simple cores designed to perform the same operation on many data items in parallel, used for graphics and other parallel tasks. Graphics processing unit (GPU)
Overlapping the fetch, decode and execute stages of successive instructions so that several are being processed at once. Pipelining	An additional processor that takes over a specific type of work from the main CPU, such as graphics or sound. Co-processor
A user interface that uses windows, icons, menus and a pointer (WIMP) to allow the user to interact visually. Graphical user interface (GUI)	The means by which a user and a computer communicate, including what is displayed and how the user gives input. User interface
A user interface that guides the user through a structured on-screen form using text boxes, drop-down lists, check boxes and radio buttons. Forms dialogue interface	A user interface in which the user types commands in an exact syntax at a prompt for the computer to carry out. Command line interface (CLI)

AS 2 · FD12

Menu-driven interface

AS 2 · FD12

Voice (dialogue) interface

AS 2 · FD12

Biometric interface

AS 2 · FD12

Natural language interface

AS 2 · FD12

Human–computer interaction (HCI)

AS 2 · FD12

Accessibility

AS 2 · FD12

Switch

AS 2 · FD13

Data compression

Backs · print double-sided, flip on long edge

A user interface that is controlled by spoken commands, using speech recognition.

Voice (dialogue) interface

A user interface that presents the user with a limited list of options to choose from, often in a hierarchy.

Menu-driven interface

A user interface that accepts requests typed or spoken in ordinary everyday language.

Natural language interface

A user interface operated by measuring a physical characteristic of the user, such as a fingerprint, face or iris.

Biometric interface

Designing systems so they can be used by people with a wide range of abilities, including visual, hearing, motor and cognitive impairments.

Accessibility

The study and design of how people interact with computers, aiming to make systems effective, efficient and satisfying to use.

Human–computer interaction (HCI)

Encoding data using fewer bits than the original representation, to reduce storage space and transmission time.

Data compression

An extra parameter typed after a command in a command line interface to modify what the command does.

Switch

AS 2 · FD13

Lossless compression

AS 2 · FD13

Lossy compression

AS 2 · FD13

Run-length encoding (RLE)

AS 2 · FD13

Dictionary-based compression

AS 2 · FD13

Compression ratio

AS 2 · FD13

Zipping

AS 2 · FD13

Colour depth

AS 2 · FD13

Bitmap (BMP)

Backs · print double-sided, flip on long edge

Compression that permanently removes data judged less important, giving smaller files that cannot be restored exactly. Lossy compression	Compression that reduces file size without losing any data, so the original can be reconstructed exactly. Lossless compression
A lossless compression method that stores repeated patterns once in a dictionary and replaces each occurrence with a short reference (token). Dictionary-based compression	A lossless compression method that replaces a run of identical values with a single value and a count. Run-length encoding (RLE)
A lossless method of compressing one or more files into a single archive file with a .zip extension. Zipping	The original file size divided by the compressed file size. Compression ratio
An uncompressed image format that stores the colour of every pixel in a grid as a binary number. Bitmap (BMP)	The number of bits used to represent the colour of each pixel; n bits give 2^n possible colours, and more bits give a larger file. Colour depth

AS 2 · FD13

GIF

AS 2 · FD13

JPEG

AS 2 · FD13

MPEG

AS 2 · FD14

Operating system

AS 2 · FD14

Kernel

AS 2 · FD14

Device driver

AS 2 · FD14

Multi-user operating system

AS 2 · FD14

Time-slicing

Backs · print double-sided, flip on long edge

An image format that uses lossy compression on 24-bit colour images, suited to photographs. JPEG	An image format using up to 256 colours from a palette chosen for the image, compressed losslessly; supports animation. GIF
System software that manages a computer's hardware and software resources and sits between the hardware, the application software and the user so that they can work together. Operating system	A family of standards for storing compressed video and audio, using lossy compression; MPEG-4 is usually stored as .mp4. MPEG
Software that allows the operating system to communicate with and control a specific hardware device. Device driver	The core of the operating system, which loads and unloads programs in memory and allocates memory, storage and processor time. Kernel
Sharing processor time by giving each task or user a short, fixed period of CPU time in turn. Time-slicing	An operating system that lets many users use one computer at the same time, sharing its resources by time-slicing while keeping each user's data private. Multi-user operating system

AS 2 · FD14

Multiprogramming

AS 2 · FD14

Multiprocessing

AS 2 · FD14

Multithreading

AS 2 · FD14

Paging

AS 2 · FD14

Segmentation

AS 2 · FD14

Interrupt

AS 2 · FD14

Round robin

AS 2 · FD14

Utility program

Backs · print double-sided, flip on long edge

Dividing work between two or more processors or cores so that it is executed truly in parallel. Multiprocessing	Holding several programs in main memory at once and switching the CPU between them, although only one executes at any instant. Multiprogramming
A memory management technique that divides memory into fixed-size pages, mapped to physical frames by a page table. Paging	Running several threads, each a separate path of execution, within one process so they share its memory and code. Multithreading
A signal to the processor from hardware or software indicating that an event needs attention; the current task is suspended while an interrupt service routine runs. Interrupt	A memory management technique that divides a program into variable-sized logical segments such as modules and data structures. Segmentation
System software that performs a specific task to help manage, maintain or protect the computer, such as compression, backup or defragmentation. Utility program	A scheduling algorithm in which each process is given a fixed time slice of the processor in turn. Round robin

AS 2 · FD14

Disk defragmenter

AS 2 · FD15

Compiler

AS 2 · FD15

Interpreter

AS 2 · FD15

Assembler

AS 2 · FD15

Intermediate code

AS 2 · FD15

Lexical analysis

AS 2 · FD15

Application software

AS 2 · FD15

Generic software

Backs · print double-sided, flip on long edge

A translator that converts a whole high-level program into object code before it is executed, producing an executable file. Compiler	A utility that rearranges the data on a magnetic hard disk so each file occupies contiguous clusters, reducing access time. Disk defragmenter
A translator that converts assembly language into machine code. Assembler	A translator that translates and executes a high-level program one statement at a time, without producing an executable file. Interpreter
The first stage of compilation, in which comments and white space are removed, the source code is converted into tokens and a symbol table is built. Lexical analysis	Processor-independent code, such as Java bytecode or C# CIL, produced by a compiler and executed by a virtual machine or JIT compiler. Intermediate code
Application software that can be used for many different tasks without modification, such as a word processor or spreadsheet. Generic software	Programs designed to let the end user carry out specific tasks, such as word processing or booking tickets, accessing the hardware through the operating system. Application software

AS 2 · FD15

Special purpose software

AS 2 · FD15

Custom-built software

AS 2 · FD15

Off-the-shelf software

AS 2 · FD15

Utility software

AS 2 · FD15

Open source software

AS 2 · FD15

Proprietary software

AS 2 · FD15

Batch processing

AS 2 · FD15

Real-time processing

Backs · print double-sided, flip on long edge

Software written specifically for one organisation's requirements, by in-house developers or a third party. Custom-built software	Application software designed to carry out one specific task or meet a particular business need. Special purpose software
System software that helps to maintain, configure and protect a computer, such as antivirus or backup software. Utility software	Ready-made software developed for a wide market and sold commercially to many customers. Off-the-shelf software
Software owned by an individual or company and distributed under a licence that keeps the source code secret and restricts copying and modification. Proprietary software	Software distributed with its source code under a licence that allows users to view, modify and redistribute it. Open source software
Processing in which each transaction or input is dealt with as soon as it occurs, so the master file is always up to date; real-time control systems also guarantee a response fast enough to affect the next input. Real-time processing	Processing in which transactions are collected over a period in a transaction file and processed together as one job at a later time, without user intervention. Batch processing

AS 2 · FD16

Internet

AS 2 · FD16

World Wide Web

AS 2 · FD16

IP address

AS 2 · FD16

Domain Name System (DNS)

AS 2 · FD16

Top-level domain (TLD)

AS 2 · FD16

URL

AS 2 · FD16

Client–server model

AS 2 · FD16

Web server

Backs · print double-sided, flip on long edge

A collection of linked web pages and resources, each with a unique address, accessed over the Internet using HTTP and browsers. World Wide Web	A global network of interconnected computer networks that use TCP/IP so that any connected device can communicate with any other. Internet
A distributed, hierarchical system of servers that translates domain names into IP addresses. Domain Name System (DNS)	A unique numeric address assigned to a device on a network such as the Internet: 32 bits in IPv4 and 128 bits in IPv6. IP address
Uniform resource locator: the full address of a resource on the Web, made up of the protocol, domain name and path or file name. URL	The last part of a domain name, such as .uk, .ie or .com, which may indicate the country or type of organisation. Top-level domain (TLD)
A computer running software that stores web pages and delivers them to browsers when requested using HTTP. Web server	A model in which client devices send requests to a central server, which processes them and returns responses. Client–server model

AS 2 · FD16

Internet service provider (ISP)

AS 2 · FD16

HTTPS

AS 2 · FD16

Cookie

AS 2 · FD16

Session

AS 2 · FD16

Cache

AS 2 · FD16

File Transfer Protocol (FTP)

AS 2 · FD17

Dynamic web page

AS 2 · FD17

Client-side scripting

Backs · print double-sided, flip on long edge

Hypertext Transfer Protocol Secure: HTTP sent over an encrypted TLS connection, with the server authenticated by a digital certificate.

HTTPS

A company that provides users with a connection to the Internet, often with services such as email and web hosting.

Internet service provider (ISP)

Data about a user's visit stored on the web server and identified by a session ID held by the browser.

Session

A small piece of text data sent by a website and stored by the browser, then returned with each request so the site can remember information about the user.

Cookie

A standard protocol for transferring files between computers, commonly used to upload website files to a web host.

File Transfer Protocol (FTP)

Temporary storage of previously retrieved web content, in the browser or on a proxy server, checked before requesting content from the web server.

Cache

Code, usually JavaScript, that is downloaded with a web page and run by the user's browser.

Client-side scripting

A web page generated at the time of request, often from database data, so its content can change or be personalised.

Dynamic web page

AS 2 · FD17

Server-side scripting

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Web application

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Native application

AS 2 · FD17

SQL injection

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E-government

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Portal

AS 2 · FD17

Intranet

AS 2 · FD17

Public key encryption

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An application that runs in a web browser and is delivered from a server, processing user input and usually storing its data in a database on the server, without being installed on the device.

Web application

Code that runs on the web server to process requests, access databases and generate pages before they are sent to the browser.

Server-side scripting

An attack in which SQL code is entered into an input field so that it is executed by the database.

SQL injection

An application written for and installed on a specific operating system, with direct access to device hardware.

Native application

A website or page that provides consistent access to a number of connected services from a single starting point.

Portal

Government information and services provided electronically, usually online, to citizens, businesses and other organisations.

E-government

Encryption using a linked key pair: data encrypted with the freely available public key can only be decrypted with the owner's secret private key.

Public key encryption

A private network, usually of web pages and online services, within an organisation that only authorised members can access.

Intranet

AS 2 · FD17

SSL/TLS

AS 2 · FD17

Digital certificate

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Digital signature

AS 2 · FD18

HTML

AS 2 · FD18

Container tag

AS 2 · FD18

Empty tag

AS 2 · FD18

Attribute

AS 2 · FD18

Anchor tag

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An electronic document, issued and signed by a certificate authority, that links a public key to the identity of its owner.

Digital certificate

Protocols that create an authenticated, encrypted link between two devices, such as a browser and a web server; TLS is the modern successor to SSL.

SSL/TLS

Hypertext Markup Language: the markup language that uses tags to define the structure and content of a web page, such as headings, paragraphs, lists, images and links.

HTML

A hash of a document encrypted with the sender's private key, used to verify who sent the document and that it has not been altered.

Digital signature

An HTML tag that stands alone with no closing tag because it encloses no content, such as
, <img> or <link>.

Empty tag

An HTML tag that is used as an opening and closing pair enclosing content, such as … or <a> … .

Container tag

The <a> tag, which creates a hyperlink; its href attribute gives the destination URL and the enclosed content is what the user clicks.

Anchor tag

Extra information written inside an HTML opening tag as name="value", which refines the element, such as href, src or alt.

Attribute

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Alt text

AS 2 · FD18

CSS

AS 2 · FD18

Selector

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External style sheet

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Inline style

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Inheritance

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Precedence

AS 2 · FD19

Target audience

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Cascading Style Sheets: the language used to describe the presentation of web pages, such as colours, fonts, spacing and layout, separately from the content. CSS	The alt attribute of an image, giving a text description that is displayed if the image cannot load and is read aloud by screen readers. Alt text
A separate .css file linked to web pages with the <link> tag so that one set of style rules controls the appearance of the whole site. External style sheet	The part of a CSS rule that identifies which HTML elements the declarations apply to, e.g. an element name, a .class or an #id. Selector
The way a nested (child) element takes on style properties such as colour and font from the element that contains it (its parent). Inheritance	A style written in the style attribute of a single HTML element, affecting only that element. Inline style
The group of people a website is designed for, described by factors such as age, interests, abilities, accessibility needs and the devices they use. Target audience	The rule that when the same property is defined in more than one place, the definition closest to the element (e.g. an inline style) takes priority. Precedence

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Site map

AS 2 · FD19

Storyboard

AS 2 · FD19

Hierarchical structure

AS 2 · FD19

Linear structure

AS 2 · FD19

WCAG

AS 2 · FD19

Responsive design

AS 2 · FD19

Web application

AS 2 · FD19

Client-side processing

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A sketch of the layout of each web page, showing the position of navigation, text, images and other elements, often with annotations. Storyboard	A diagram showing all the pages of a website and how they are linked. Site map
A navigation structure in which pages are visited in a fixed sequence using next and previous links. Linear structure	A navigation structure in which the home page links to main sections, which in turn link to their own sub-pages. Hierarchical structure
Web design in which the layout adapts automatically to the size of the screen being used. Responsive design	Web Content Accessibility Guidelines: the international standard for accessible web content, based on the principles perceivable, operable, understandable and robust. WCAG
Processing carried out by the browser on the user's device, such as validating form input or changing the page, without contacting the server. Client-side processing	An application that runs in a web browser and is delivered from a server, processing user input and usually storing its data in a database on the server, without being installed on the device. Web application

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Server-side processing

AS 2 · FD19

Broken link

AS 2 · FD19

Web hosting

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A hyperlink whose destination does not exist, producing an error such as 404 Not Found. Broken link	Processing carried out on the web server after a request is submitted, such as querying and updating a database and generating the response page. Server-side processing
	A service that stores a website's files on servers permanently connected to the internet so the site can be accessed by users. Web hosting